

ASSIGNMENT -02

**Take one Domain and draw the graph (Normal distribution)
(Empirical rule)**

Introduction

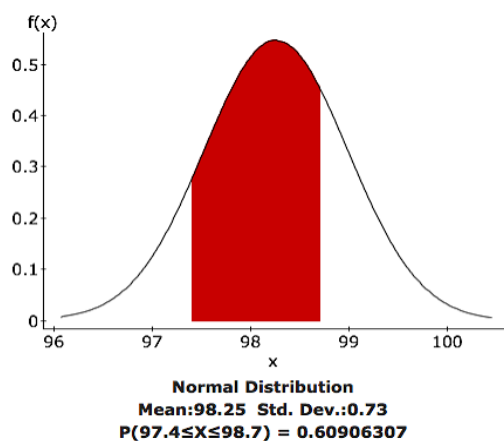
Statistics plays a crucial role in understanding patterns in real-world data. One of the most important probability distributions in statistics is the **Normal Distribution**, also known as the bell curve. Many biological measurements naturally follow this distribution.

In this report, we choose the domain of **Human Body Temperature** to explain:

- Normal Distribution
- Standard deviation
- Graphical representation
- Empirical Rule (68–95–99.7 Rule)
- Practical interpretation

Body temperature measurements in a large healthy population approximately follow a normal distribution.

Understanding the Domain – Body Temperature



Medical studies show:

- Average normal body temperature $\approx 98.6^\circ\text{F}$
- Slight variation exists among individuals.

Assume:

Mean $(\mu) = 98.6^\circ\text{F}$
 Standard Deviation $(\sigma) = 0.7^\circ\text{F}$

Most people have temperatures close to 98.6°F , while very high or very low temperatures are rare.

What is Normal Distribution?

A Normal distribution has the following properties:

- Symmetrical bell-shaped curve
- Mean = Median = Mode
- Total probability area = 1
- Defined by mean (μ) and standard deviation (σ)

Mathematical Formula

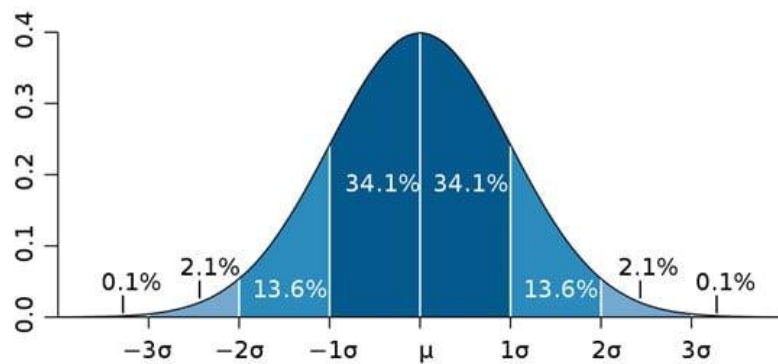
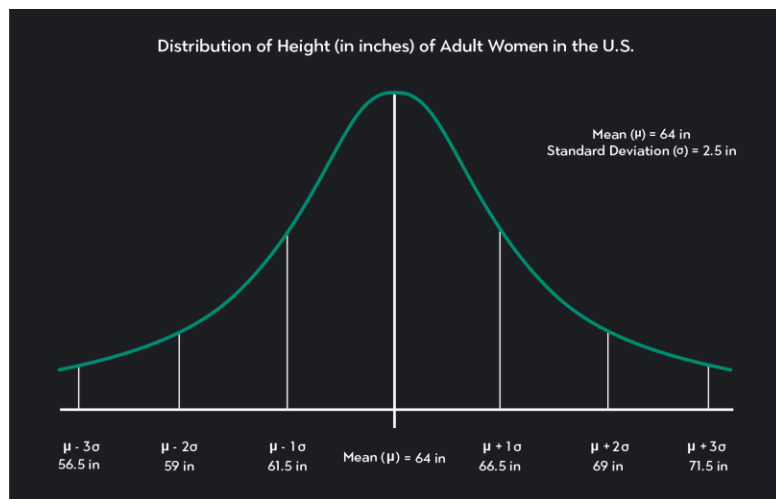
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = Mean
- σ = Standard deviation

This function describes how data values are distributed around the mean.

Graph of Normal Distribution (Body Temperature)



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Graph Description:

- Center = 98.6°F
- Curve spreads equally on both sides
- Most people cluster near the mean

Few people:

- Below 97°F
- Above 100°F

These are extreme cases.

Standard Deviation and Spread

Standard deviation measures variability.

If σ is small:

- Temperatures are tightly clustered
- Curve is narrow

If σ is large:

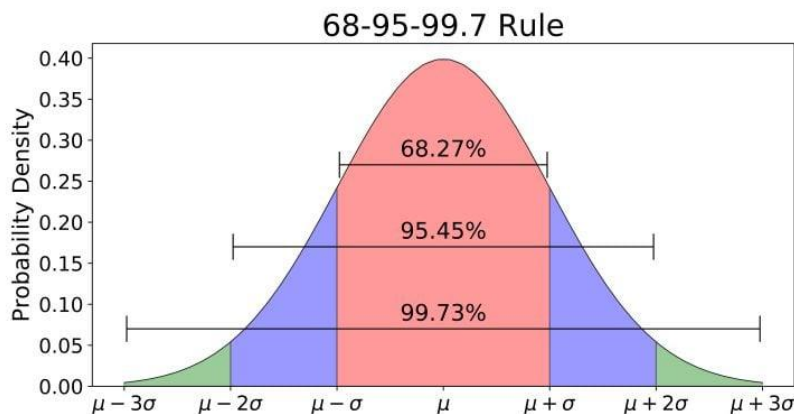
- Greater variability
- Curve is wider

In our example:

$$\sigma = 0.7^{\circ}\text{F}$$

This means temperatures typically vary by about $\pm 0.7^{\circ}\text{F}$ from the average.

Empirical Rule (68–95–99.7 Rule)



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The Empirical Rule states:

- 68% of data lies within 1σ
- 95% lies within 2σ
- 99.7% lies within 3σ

This rule applies only to normally distributed data.

Applying Empirical Rule to Body Temperature

Given:

Mean (μ) = 98.6°F
Standard Deviation (σ) = 0.7°F

1 Standard Deviation ($\pm 1\sigma$)

$$98.6 \pm 0.7$$

Range: 97.9°F to 99.3°F

About 68% of healthy people fall in this range.

2 Standard Deviations ($\pm 2\sigma$)

$$98.6 \pm 1.4$$

Range: 97.2°F to 100.0°F

About 95% of healthy people fall in this range.

3 Standard Deviations ($\pm 3\sigma$)

$$98.6 \pm 2.1$$

Range: 96.5°F to 100.7°F

About 99.7% of healthy individuals fall within this range.

Very few fall outside this.

Interpretation in Medical Context

Doctors use this concept to:

- Detect fever
- Identify abnormal conditions
- Analyze population health data

If temperature exceeds 100.7°F:

- It lies outside 3σ
- Considered statistically rare
- May indicate infection

Normal distribution helps identify abnormal medical conditions.

Importance of Normal Distribution in Healthcare

Normal distribution helps in:

- Clinical trials
- Medical research
- Setting diagnostic thresholds
- Identifying outliers
- Risk prediction

Many biological variables follow approximately normal distribution:

- Blood pressure
- Cholesterol levels
- Heart rate

Conclusion

Human body temperature is an excellent real-world example of Normal Distribution. Most individuals have temperatures near the average, while extreme temperatures are rare.

Using the Empirical Rule:

- 68% between 97.9°F and 99.3°F
- 95% between 97.2°F and 100.0°F
- 99.7% between 96.5°F and 100.7°F

Normal distribution provides a powerful framework for understanding variability in biological measurements. It allows healthcare professionals to distinguish between normal variation and abnormal medical conditions.

Thus, the concepts of Normal Distribution and Empirical Rule are not only theoretical but have direct practical applications in medicine and public health.